The Autocorrelation Mirage: Pseudoreplication Dominates Leakage in Rare-Event Panel Forecasting

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The Autocorrelation Mirage: Pseudoreplication Dominates Leakage in Rare-Event Panel Forecasting

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Rare-event forecasting from panel data faces a sample-size barrier: too few episodes for stable inference. A common solution expands episodes into daily observations, but this activates a failure mode we call *episode memorization*: models learn to identify which episode an observation belongs to rather than predicting event risk. We show that standard K-Fold cross-validation—the default in libraries like Scikit-Learn—enables this failure by allowing the same episode to appear in both training and test sets. The result is dramatic: AUC inflates from 0.56 to 0.86 (+54%) with *no data leakage whatsoever*. This exceeds even explicit time-to-event leakage (+39%). While prior work formalized leakage for i.i.d. data (Kaufman et al.) and block CV for spatial dependence (Roberts et al.), neither identified that *pseudoreplication alone* dominates in panel-expanded forecasting. We propose the Δ_{CV} diagnostic: the gap between random and grouped CV estimates. In our simulation, $\Delta_{\text{CV}} = 0.30$—a gap this large indicates the model exploits autocorrelation structure rather than learning generalizable signal. This pathology affects conflict prediction, clinical deterioration, equipment failure, and customer churn systematically.

Additional Key Words and Phrases: pseudoreplication, cross-validation, panel data, rare events, episode memorization

1 Introduction

Predicting rare events—coups, equipment failures, patient deterioration—from panel data confronts a sample-size barrier. With 20–50 episodes and 10–20 features, events-per-variable (EPV) ratios fall below 1.0, far short of the 10–20 required for stable inference [7, 9]. A common solution expands each episode into daily rows: 20 episodes become 500+ observations, apparently resolving the problem.

We show this reformulation activates a failure mode we call **episode memorization**. When standard K-Fold cross-validation—the default in Scikit-Learn and similar libraries—assigns observations to folds, the same episode appears in both training and test sets. Because consecutive observations $X_{e,t}$ and $X_{e,t+1}$ are highly autocorrelated, the model effectively performs nearest-neighbor classification: it learns to identify the episode's identity ("this is the high-variance episode") rather than the event risk. At test time, recognizing familiar autocorrelation patterns yields artificial accuracy. Random CV thus transforms a *forecasting* problem (extrapolation to unseen episodes) into an *interpolation* problem (pattern-matching within known episodes).

In our simulation, this single methodological choice—`KFold` instead of `GroupKFold`—inflates AUC from 0.56 to 0.86. This 54% inflation occurs with strictly causal features containing *no leakage whatsoever*. Remarkably, this exceeds explicit time-to-event leakage (+39%), where features directly encode $T_e - t$. This finding reframes the rare-event forecasting literature. Kaufman et al. [4] formalized data leakage and "learn-predict separation," but focused on i.i.d. settings with explicit feature violations. Roberts et al. [8] established that block CV is "nearly universally more appropriate" for structured data, but addressed spatial dependence in ecology. Hurlbert [2] coined "pseudoreplication"—finding it in 48% of ecological studies—but addressed experimental design, not ML evaluation. We show that panel expansion creates a setting where pseudoreplication *dominates* leakage: the standard tool is more dangerous than the worst feature-engineering error.

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Table 1. Common operations and predictability violations.

Operation	Standard (Violates)	Corrected
Normalization	Global (μ, σ)	Rolling/fold-specific
Imputation	Column/episode mean	Forward-fill only
Moving average	Centered window	Trailing window
Lag features	Include $t, t{+}1$	Strictly $s < t$

Kapoor and Narayanan [3] documented leakage in 294+ papers across 17 fields; Neunhoeffer and Sternberg [6] showed CV misuse invalidated civil war prediction claims. We extend this by: (1) demonstrating pseudoreplication as the dominant failure mode; (2) formalizing episode memorization as the mechanism; (3) proposing the Δ_{CV} diagnostic; and (4) providing a corrected protocol.

2 Theoretical Framework

Consider E episodes indexed by e. Each has time $t \in \{0, \ldots, T_{\max}\}$, event time T_e, features $X_{e,t} \in \mathbb{R}^p$, and static attributes α_e. For horizon H, the target is $Y_{e,t}(H) := \mathbb{1}\{t < T_e \leq t + H\}$, evaluated when at-risk ($t < T_e$).

Let $\mathcal{F}_{e,t^-} := \sigma(\{X_{e,s} : s < t\} \cup \{\alpha_e\} \cup \mathcal{E}_{<t})$ denote information strictly before t.

Definition 1 (Predictable Feature). *Feature $X_{e,t}$ is* predictable *if $X_{e,t} \in \mathcal{F}_{e,t^-}$—measurable with respect to information available before t.*

This extends Kaufman et al.'s [4] learn-predict separation to filtrations. Violations need not be explicit: global statistics computed over $\{X_{e,s} : s \leq T_e\}$ implicitly depend on T_e.

Theorem 1 (Leakage Guarantees Inflation). *If $X_{e,t} = f(T_e - t) + \epsilon_{e,t}$ with f monotone and $\text{Var}(\epsilon) = \sigma^2$, then $\lim_{\sigma^2 \to 0} AUC = 1$.*

Proof sketch. Since $Y_{e,t}(H) = 1$ iff $T_e - t \in (0, H]$, monotone f creates a threshold classifier that perfectly separates as noise vanishes. □

3 Implicit Violations in Standard Pipelines

Global normalization $\tilde{X} = (X - \mu)/\sigma$ uses statistics from all observations. When features exhibit within-episode trends toward the event, μ and σ depend on $\{T_e\}$, creating implicit leakage. *Correction:* fold-specific or rolling statistics. Mean imputation uses future observations; episode-mean imputation uses $\bar{X}_e$ which depends on T_e. *Correction:* forward-fill only. Centered moving averages use future values. *Correction:* trailing windows only.

These traps require features with pre-existing trends toward the event. Strictly causal features (constructed only from past values) are immune. Real-world features often exhibit such trends—vital signs deteriorating before death, SMART errors accumulating before drive failure, conflict indicators escalating before war—making these traps practical concerns even when absent in synthetic benchmarks.

4 Episode Memorization: The Dominant Failure

Even with leak-free features, standard K-Fold CV inflates performance through *episode memorization*. This is Hurlbert's [2] pseudoreplication translated to ML: "replicates are not statistically independent." Within-episode observations share: (1) latent episode effects α_e; (2) autocorrelated features with $\text{Corr}(X_{e,t}, X_{e,t+1}) \gg 0$; (3) labels determined by the same T_e. When random CV places observations from episode e in both training and test sets, the model learns episode-specific

Table 2. Simulation results (100 replicates). Pseudoreplication dominates leakage.

Condition	AUC	Brier	Inflation
Leak-Free + Grouped CV	0.56 ± 0.07	0.29 ± 0.05	baseline
Leak-Free + Random CV	$\mathbf{0.86 \pm 0.04}$	0.15 ± 0.02	**+54%**
Norm. Leak + Grouped	0.56 ± 0.07	0.29 ± 0.05	+0%[†]
Explicit Leak + Grouped	0.77 ± 0.04	0.16 ± 0.03	+39%

[†]See text for interpretation.

patterns—the "texture" of each episode's autocorrelation structure. At test time, it performs *identity inference*: recognizing which episode an observation belongs to, then recalling the associated outcome. This transforms forecasting (extrapolation) into interpolation (pattern-matching). Under intraclass correlation ρ with m observations per episode: $n_{\text{eff}} = \frac{n}{1+(m-1)\rho}$.

For $n = 600$, $m = 20$, $\rho = 0.5$: $n_{\text{eff}} = 55$—a 10× reduction. But the inflation from episode memorization is *not* merely a sample-size effect: even with $n_{\text{eff}} < 20$, random CV achieves AUC > 0.8 (Figure 1D).

Proposition 1. *Under within-episode dependence ($\rho > 0$): $\mathbb{E}[\widehat{AUC}_{random}] > \mathbb{E}[\widehat{AUC}_{grouped}]$, with gap increasing in ρ and model capacity.*

5 Empirical Validation

We generate $E = 30$ episodes with: episode effect $\alpha_e \sim \mathcal{N}(0, 0.5)$; AR(1) latent process $Z_{e,t} = 0.7Z_{e,t-1} + \epsilon_t$; hazard $h_{e,t} = \text{logit}^{-1}(-3 + \alpha_e + 0.15Z_{e,t})$; features from $Z_{e,<t}$ only (strictly causal); label $Y_{e,t} = \mathbb{1}\{t < T_e \leq t + 14\}$. The intercept -3 yields an approximate 5% base event rate ($\text{logit}^{-1}(-3) \approx 0.047$), matching empirical rare-event frequencies. Features are constructed only from $Z_{e,<t}$ (strictly causal); the label is $Y_{e,t} = \mathbb{1}\{\, t < T_e \leq t + 14 \,\}$. We evaluate four conditions across 100 replicates.

Table 2 and Figure 1 reveal the central finding: pseudoreplication alone inflates AUC more than explicit leakage. Random CV yields AUC=0.86 from features with no leakage—a 54% inflation over grouped CV (AUC=0.56). The Δ_{CV} gap is 0.30. Explicit $T_e - t$ leakage inflates by only 39%. Global normalization shows +0% inflation because our synthetic features are stationary AR(1) processes with no pre-event trends. This is a *conservative* design choice, not a refutation of the normalization trap. In real-world domains with pre-event trends (deteriorating vitals, accumulating errors), the normalization trap activates. The critical finding is that pseudoreplication inflates AUC to 0.86 *even in this trend-free setting*—autocorrelation alone is sufficient to destroy validity.

Figure 1D demonstrates that under random CV, performance completely decouples from effective sample size. The model achieves high AUC regardless of n_{eff}, indicating it exploits episode texture rather than learning generalizable signal.

6 The Δ_{CV} Diagnostic

We propose a simple diagnostic for episode memorization:

Definition 2 (Δ_{CV} Diagnostic). *$\Delta_{CV} := \widehat{AUC}_{random} - \widehat{AUC}_{grouped}$. A gap $\Delta_{CV} > 0.05$ indicates the model relies on episode memorization rather than generalizable signal.*

In our simulation, $\Delta_{\text{CV}} = 0.30$. This metric requires no additional data collection—only re-running evaluation with `GroupKFold` instead of `KFold`.

Corrected Protocol: (1) *Preprocessing:* Fit transformations on training fold only; forward-fill imputation; trailing windows. (2) *Evaluation:* Use `GroupKFold` with `groups=episode_id`; never split episodes across folds. (3) *Reporting:* Report Δ_{CV} and n_{eff} alongside AUC.

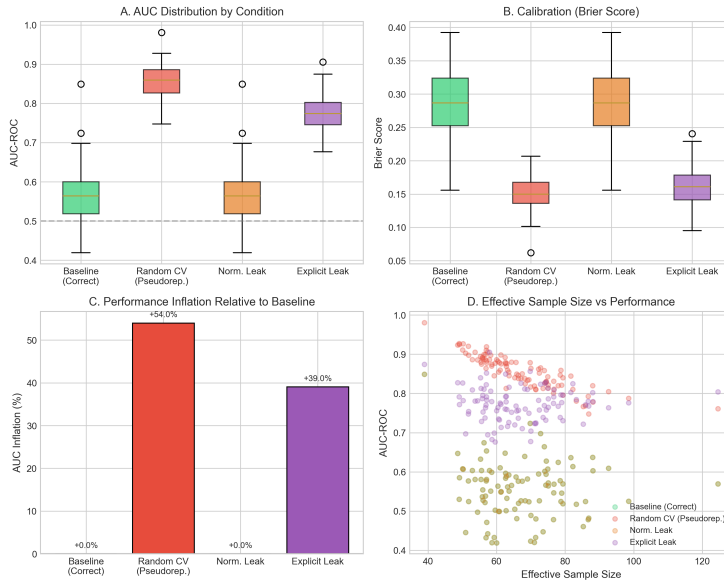

Fig. 1. Simulation results (100 replicates). *(A)* AUC distributions: random CV inflates performance dramatically. *(B)* Brier scores confirm improved calibration under random CV is spurious. *(C)* Pseudoreplication (+54%) exceeds explicit leakage (+39%)—the standard tool is more dangerous than the worst feature error. *(D)* Under random CV (red), performance completely decouples from effective sample size: the model achieves AUC > 0.8 even when $n_{\text{eff}} < 50$, indicating exploitation of autocorrelation artifact rather than learned signal.

Red flags: AUC > 0.85 with $E < 50$ episodes; $\Delta_{\text{CV}} > 0.10$; Brier < 0.10 on imbalanced data.

7 Discussion

Leakage requires specific conditions: explicit $T_e - t$ features, or trending features combined with global normalization. Pseudoreplication requires only default `KFold`—a single line of code that every practitioner uses. This asymmetry explains why pseudoreplication is the greater threat: it is invisible, ubiquitous, and activated by default.

We unify Kaufman et al. [4] (leakage formalization), Roberts et al. [8] (block CV), and Hurlbert [2] (pseudoreplication). The contribution is demonstrating that pseudoreplication *dominates* in panel-expanded forecasting and providing the Δ_{CV} diagnostic. The pathology applies wherever rare events are expanded into panels: conflict forecasting [1, 5], clinical deterioration, predictive maintenance, customer churn. Our DGP uses stationary features; real-world trending features would activate the normalization trap, potentially compounding inflation beyond 54%. We focus on gradient boosting; sequence models may have additional vulnerabilities through attention-based episode memorization.

Panel expansion creates value only if evaluation respects episode boundaries. Standard K-Fold CV—the default in Scikit-Learn—violates this, enabling episode memorization that inflates AUC by 54%. This exceeds explicit time-to-event leakage (+39%), making the standard tool more dangerous than the worst feature-engineering error. The Δ_{CV} diagnostic (random minus grouped CV) provides a simple check: our simulation shows $\Delta_{\text{CV}} = 0.30$. We call for retrospective audits of published rare-event forecasting claims and recommend that reviewers require grouped CV, Δ_{CV} reporting, and n_{eff} disclosure.

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